

SUMMER TRAINING REPORT

on

Automating Learning Management System

TRAINING DETAILS

Stratbeans Consulting (Remote)

TRAINING INCHARGE

Mr. Sapan Kumar, Senior Manager – HR & Admin

PERIOD

01/06/2026 to 24/07/2026

SUBMITTED BY

Yash Agarwal (23119039)



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

JAYPEE INSTITUTE OF INFORMATION TECHNOLOGY,

NOIDA (U.P.) AUGUST 2026

CERTIFICATE



Stratbeans Consulting Pvt. Ltd.
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www.stratbeans.com

HR/ Intern/Certificate/2707/26/01

Date: 27th July 2026

To whomsoever it may concern

This is to certify that Mr. Yash Agarwal was part of an internship project as **“Software Development Engineer - Intern”** at Stratbeans Consulting Pvt. Ltd. from **01st June 2026 to 24th July 2026.**

During his tenure with us, we found him to be quite hardworking, sincere and result oriented. We wish him all success in his future endeavors and career.

For Stratbeans Consulting Pvt. Ltd.



Sapan Kumar
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DECLARATION

I, **Yash Agarwal**, hereby declare that this summer training report entitled "**Automating Learning Management System**" submitted to **Jaypee Institute of Information Technology, Noida**, for the partial fulfilment of the requirements of the bachelor of technology is the bonafide record of the work done by me in the due course of my summer training and the contents and facts prepared and presented by me without any bias and are authentic to the best of my knowledge. I also declare that it has not previously formed the basis of an award for me any degree/ diploma, fellowship or other similar titles of any institute/ society.

Name: **YASH AGARWAL**

Enrollment Number: 23119039

Date:

ACKNOWLEDGEMENT

The time I spent in **Stratbeans Consulting** as a trainee from **1st June 2026** to **24th July 2026** was a memorable one for me as it was rich in experience sharing and helped me to discover my potential. I have had so many experiences and opportunities that I believe will forever shape and influence my professional skills while fostering personal growth and development.

With a sage sense of gratitude, I acknowledge various people who directly or indirectly contributed to the development of this work and who influenced my thinking, behavior & acts during training. I express my deep sense of gratitude to **Mr. Sapan Kumar** for his technical support and constant supervision which contributed immensely to my development. I also thank him for his guidance which was a remarkable force that enabled me to complete the training program.

I would like to thank my parents and my friends whose love and guidance are with me in whatever I pursue. Nobody has been more important to me than the members of my family and friends.

I also express my sincere gratitude to **T&P Team**, Jaypee Institute of Information and Technology, for their stimulating guidance, continuous encouragement and evaluating me on this training

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Chapter 1 Introduction

1.1 Background of Summer Training

Summer training is an important part of engineering education because it exposes students to real industrial practices. In a classroom environment, concepts such as programming, databases, software engineering, web technologies, and user interface design are usually studied separately. During an internship, the same concepts come together in the form of a working product. This helps a student understand not only how software is written, but also why a particular feature is needed and how it affects users.

My summer training was carried out at Stratbeans Consulting, an organization working in the area of digital learning and learning management solutions. The training was focused on understanding and contributing to a Learning Management System (LMS). An LMS is a software platform that helps organizations manage courses, learner enrollment, progress tracking, learning analytics, and digital content delivery. Such systems are widely used by educational institutions, companies, and training departments for structured learning.

During the internship, I worked as a trainee on different parts of the LMS platform. The work included environment setup, understanding the existing codebase, frontend interface improvements, backend study using PHP, user management workflows, course enrollment, progress tracking, analytics dashboard concepts, recommendation logic, and a learning assistant chatbot. The work was not limited to writing code only, it also involved reading documentation, discussing requirements, testing pages on different screens, and preparing notes for future reference.

1.2 About Stratbeans Consulting

Stratbeans Consulting develops digital learning products and services for organizations that need structured training solutions. The company works in areas such as learning management systems, digital content, learner engagement, assessment support, analytics, and enterprise learning. Its products are designed to help clients deliver learning material, monitor learner progress, and improve the overall training experience.

The LMS platform studied during the internship included learner-facing modules as well as administrative features. Learners can access courses, view progress, receive recommendations, and move through learning content. Administrators can manage users, courses, enrollments, and reports. This combination of frontend, backend, and database-driven functionality made the internship useful from a software engineering point of view.

1.3 Motivation of the Work

The main motivation behind the internship was to understand how a software product is developed and maintained in an industry environment. Learning platforms are important in the current industry scenario because organizations need scalable ways to train employees, update skills, and track learning outcomes. A well-designed LMS reduces manual effort and helps users learn at their own pace while giving administrator's useful information about participation and performance.

Another motivation was to improve practical development skills. Before the internship, most of my programming experience was based on academic assignments or small projects. Working on an existing LMS introduced me to a larger codebase, coding standards, Git workflows, responsive design, and the need for proper documentation. These are skills that are difficult to learn fully without practical exposure.

1.4 Objectives of the Training

The major objective of the training was to understand the structure and working of a real LMS platform and contribute to selected modules under guidance. The training aimed to improve my understanding of web application development, especially the connection between frontend screens, backend logic, and database records. It also aimed to develop professional habits such as documenting work, taking feedback, and testing implemented changes.

Specific objectives included setting up the required development environment, learning the PHP-based project structure, understanding learner and administrator workflows, improving responsive design of selected screens, studying user management and enrollment modules, contributing to progress tracking and analytics-related features, exploring recommendation logic, and planning chatbot support for common learner queries.

1.5 Scope of the Report

This report covers the complete training work from induction to final documentation. It explains the basic technologies used, the literature and background of LMS platforms, the methodology followed during development, the modules studied, the implementation activities performed, and the final results and learning outcomes. The report is written from the perspective of a third-year engineering student and therefore explains concepts in a simple and clear manner without unnecessary jargon.

Chapter 2 Basics of the Work with Literature Survey

2.1 Learning Management System: Basic Concept

A Learning Management System is a software application used to plan, deliver, manage, and track learning activities. It provides a common platform where learners can access courses and administrators can manage learning content. In many organizations, an LMS acts as the central system for training employees, onboarding new members, conducting assessments, and measuring learning progress.

The core idea of an LMS is to make learning organized and measurable. Instead of sharing course material manually, the platform stores content in a structured form. Learners can log in, view assigned courses, complete activities, and track their own progress. At the same time, administrators can see enrollment details, completion reports, and engagement metrics. This makes the system useful for both learners and training managers.

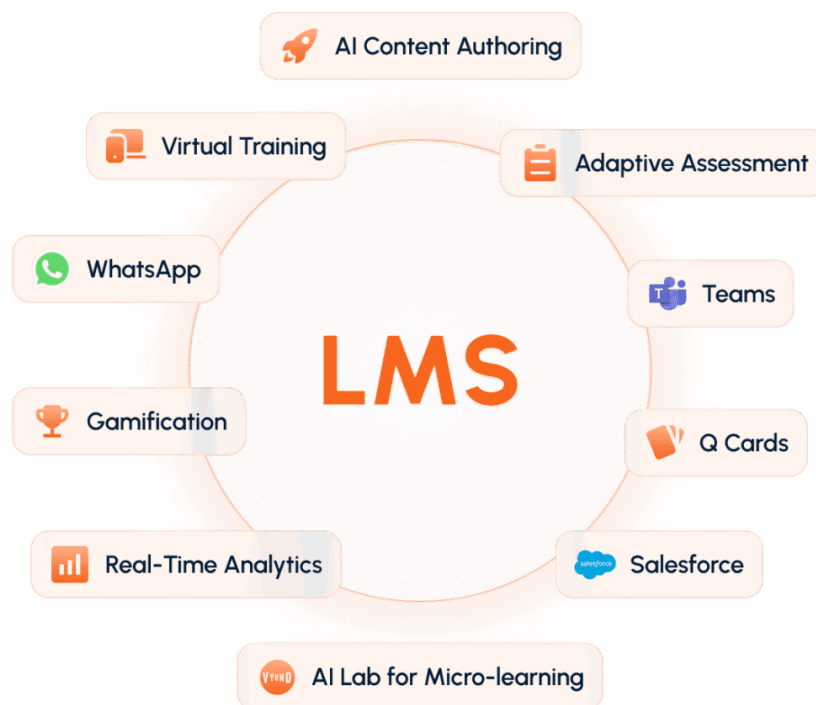


Fig 2.1: Learning Management System

2.2 Literature Survey and Industry Context

The use of web-based learning platforms has increased due to the growth of online education, remote work, and continuous professional development. Industry reports and technical literature describe digital learning as a major part of modern workforce development. Organizations need employees to learn new tools and processes quickly, and LMS platforms provide a structured way to deliver that learning.

Research and industry practice show that learning platforms are more effective when they include progress tracking, analytics, personalized suggestions, and easy navigation. A learner should be able to understand what course to take, how much work is pending, and where to continue from. Administrators need reliable data to understand participation and course effectiveness. Therefore, features such as dashboards, recommendation systems, and chat-based support have become important in modern LMS products.

From a software engineering point of view, an LMS is a good example of a multi-module web application. It includes authentication, user roles, content management, database operations, analytics, and user interface design. Literature on software engineering emphasizes modular design, maintainability, usability, and iterative development. These ideas were visible in the project because each module had to work with the rest of the system while still remaining understandable for future developers.

2.3 Technologies Used

The internship involved working with common web development tools and technologies. Visual Studio Code was used as the main code editor because it supports extensions, formatting, and easy navigation of project files. Git was used for version control and repository management. XAMPP was used to run a local server environment with PHP support. PHP was used for backend development and processing dynamic requests. Frontend changes involved HTML, CSS, and related components used in the existing project.



Fig 2.2: Major Technologies Used

2.4 Software Engineering Concepts Applied

Several software engineering concepts were observed during the internship. The first was modularity. The LMS was divided into different modules such as user management, course management, enrollment, progress tracking, analytics, recommendation, and chatbot support. This division made it easier to understand the system and work on selected features without disturbing unrelated areas.

The second concept was maintainability. In a real project, code is not written for one-time use only. Other developers may need to read, modify, or extend it later. Therefore, I learned the importance of following existing coding patterns, keeping changes readable, and documenting important workflows. Even small frontend or backend changes were discussed in relation to their effect on the whole product.

The third concept was usability. A technically correct feature may still be weak if users find it difficult to understand. While working on dashboard and portal screens, I learned that spacing, alignment, responsiveness, and clear navigation can strongly affect the learner experience. This helped me understand the connection between software development and user satisfaction.

2.5 Importance in Current Industry Scenario

The area of digital learning is important because the need for continuous upskilling has increased across industries. Traditional classroom training alone is not always flexible enough for large organizations. LMS platforms solve this problem by making learning accessible, trackable, and scalable.

Features such as analytics and recommendations add further value. Analytics help training teams identify whether learners are completing courses and where they may be facing difficulty. Recommendations help learners discover relevant content based on their learning journey. Chatbot support can reduce the time spent searching for information.

Chapter 3 Project Understanding and Methodology

3.1 Initial Induction and Environment Setup

The first phase of the internship involved induction and environment setup. I was introduced to Stratbeans Consulting, its learning solutions, and the purpose of the LMS platform. This was important because understanding the business problem helped me connect the technical tasks with actual user needs. I installed Visual Studio Code, Git, XAMPP, PHP, and the required extensions for development.

After installation, I verified that the local server and development tools were working correctly. I also received access to project repositories and internal resources. The setup process taught me that a stable development environment is the first requirement for productive work. Small issues in configuration can delay later tasks, so I documented important setup steps and tool details for future reference.

3.2 Understanding the Existing Codebase

After the setup phase, I focused on understanding the existing codebase. The team explained the architecture of the LMS platform through knowledge transfer sessions. I explored project folders, reviewed PHP files, and observed how frontend and backend components were organized. This helped me become familiar with naming patterns, reusable components, utility functions, and shared modules used in the project.

I also studied how learner accounts, course management, enrollment processes, and progress tracking were connected. Understanding these relationships was important before making any changes. A change in one module can affect another module if the data flow is not understood properly. Therefore, I maintained notes and simple diagrams to record how different modules interacted with one another.

3.3 Methodology Followed

The methodology followed during the internship was iterative and guided. For each task, I first tried to understand the requirement and the relevant part of the existing codebase. After that, I studied similar implementations already present in the project. This helped me follow the team style and avoid unnecessary changes. Once the approach was clear, I made small changes, tested them locally, and discussed the outcome with my mentor.

For frontend tasks, the methodology involved reviewing the current screen, identifying layout or usability issues, applying CSS or component-level improvements, and testing the result on different screen sizes. For backend tasks, the method involved reading the workflow, understanding how form data or user activity was processed, and reviewing database interactions. Documentation was updated after each significant task so that the learning could be used later.

3.4 Learner Journey in the LMS

A major part of the project understanding was related to the learner journey. The journey starts when a user logs in and reaches the learner dashboard. From the dashboard, the learner can view enrolled courses, continue learning, check progress, and access recommendations or support. Each step depends on proper backend data and a clear frontend interface.



Fig 3.1: Different Features of Learning Management System

3.5 Documentation Practice

Documentation was maintained throughout the internship. I prepared notes about setup steps, module structure, workflows, and implementation observations. This practice helped me revise concepts later and also improved my ability to explain technical work in an organized way. Documentation is important in industry because it reduces dependency on individual memory and helps new team members understand previous work faster.

Chapter 4 Frontend Development and UI Enhancements

4.1 Purpose of Frontend Work

Frontend development was one of the first active contribution areas during the internship. The purpose of the frontend work was to improve the usability and responsiveness of selected LMS screens. A learning platform is used by learners from different devices, so pages should display properly on desktop, tablet, and mobile screens. If important information is hidden, misaligned, or difficult to read, the learning experience becomes weaker.

My frontend tasks included reviewing learner-facing screens, identifying layout improvements, adjusting alignment, improving readability, refining dashboard sections, and testing responsiveness. These tasks helped me understand that frontend development is not only about appearance. It also affects navigation, clarity, accessibility, and the speed with which a user can complete a task.

4.2 Interactive Learner Dashboard

The Interactive Learner Dashboard was an important area of work. I studied how learners interact with dashboard sections and how they access ongoing courses, progress information, and important actions. The objective was to organize information so that learners could quickly understand their current learning status. Dashboard sections were reviewed to make course details and progress indicators easier to notice.

Additional widgets were refined to present important information in a cleaner manner. Navigation pathways between modules were also studied so that frequently used features could be accessed with fewer steps. This work showed me that dashboard design requires a balance between showing enough information and avoiding unnecessary clutter. A dashboard should support quick decisions, not confuse the user.

4.3 Course Management Portal Screens

I also contributed to frontend improvements in the Course Management Portal. These screens are used for managing learning content and course-related information. The work involved reviewing form layouts, improving the flow of content creation and editing, and ensuring that actions were easy to understand. Forms are a critical part of administrative workflows because small layout issues can slow down repeated tasks.

Particular attention was given to consistency across screens. If similar fields or actions appear differently on different pages, users may hesitate or make mistakes. Therefore, I observed existing patterns and tried to keep improvements aligned with them. The work strengthened my understanding of interface design for complex workflows.



Fig 4.1: Leadership Board for Courses

4.4 Responsiveness and Visual Consistency

Responsiveness was tested by checking how pages behaved on different screen widths. In some cases, content required better spacing or alignment to remain readable. I learned that responsive design must be planned carefully because a layout that works well on a large screen may not automatically work on a smaller one. CSS adjustments were made to improve page organization and reduce display issues.

Visual consistency was another important part of the work. The LMS includes multiple modules, but users should feel that they are using one connected product. Consistent spacing, fonts, colors, and component behavior help build that feeling. While making frontend improvements, I tried to understand the existing design language and make changes that supported it.

4.5 Learning from Frontend Tasks

The frontend tasks helped me understand the importance of user-centered thinking. A developer may know where a button or feature exists, but a learner or administrator should be able to find it naturally. Interface improvements should therefore be based on actual use cases. I also learned that even small changes such as spacing, alignment, or clearer grouping can improve the overall experience of the platform.

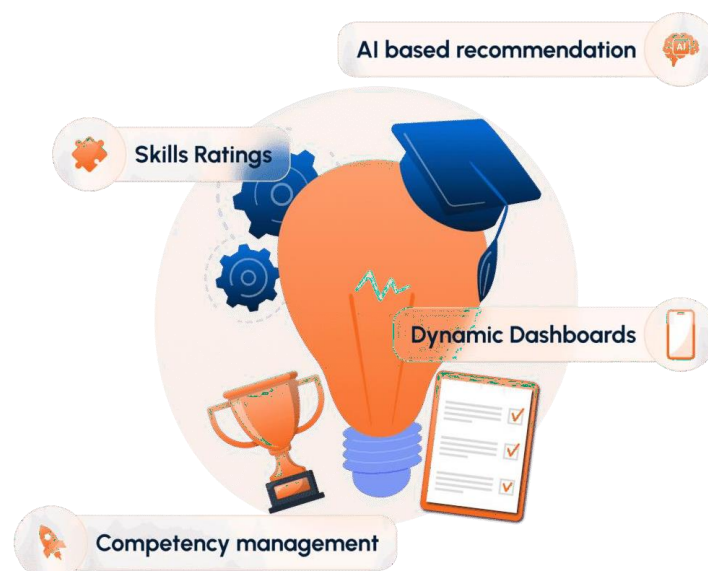


Fig 4.2: Learning Experience Platform

Chapter 5 Backend Modules and Feature Development

5.1 Introduction to Backend Work

After gaining confidence with the frontend and project structure, I moved toward backend-related learning and development tasks. The backend of the LMS was mainly studied through PHP modules and database-driven workflows. Backend logic is responsible for processing user requests, validating data, updating records, and sending required information to frontend pages. This part of the internship helped me connect programming concepts with real application behavior.

5.2 User Management Module

The User Management Module was one of the first backend areas I studied. I reviewed how user information is stored, processed, and updated in the LMS. This included understanding registration workflows, profile management, form validation, and secure data handling practices. User management is a basic but very important module because almost every other feature depends on correct user identity and role information.

I also explored role-based access control. Different users may have different permissions depending on whether they are learners, administrators, trainers, or other user categories. Studying this module helped me understand why authentication and authorization are important in enterprise applications. It also taught me that backend code must be careful while handling user data because errors can create security or access issues.

5.3 Course Enrollment System

The Course Enrollment System manages the connection between learners and courses. I studied how learners are enrolled into courses and how enrollment records are maintained. This required understanding database relationships between user records, course records, and enrollment status. The module also introduced me to edge cases such as duplicate enrollment, inactive courses, or different enrollment scenarios.

Working on this module improved my understanding of backend workflows. A single action on the frontend, such as enrolling in a course, may trigger multiple backend checks and database updates. The experience showed me that backend development requires careful logical thinking because the system must handle normal and exceptional cases properly.

5.4 Learning Progress Tracking

Progress tracking is one of the most useful parts of an LMS. It records learner activity and helps calculate completion percentages. I reviewed how progress information is updated as learners complete different activities. This information is then displayed on dashboard screens and may also be used in analytics reports. Accurate progress tracking is important because learners and administrators both depend on it.

The progress tracking work helped me understand how backend systems support meaningful user-facing features. A progress bar or completion percentage may look simple on the frontend, but it depends on correct activity records, course structure, and calculation logic. I documented the workflow so that the connection between activity, progress update, and dashboard display remained clear.

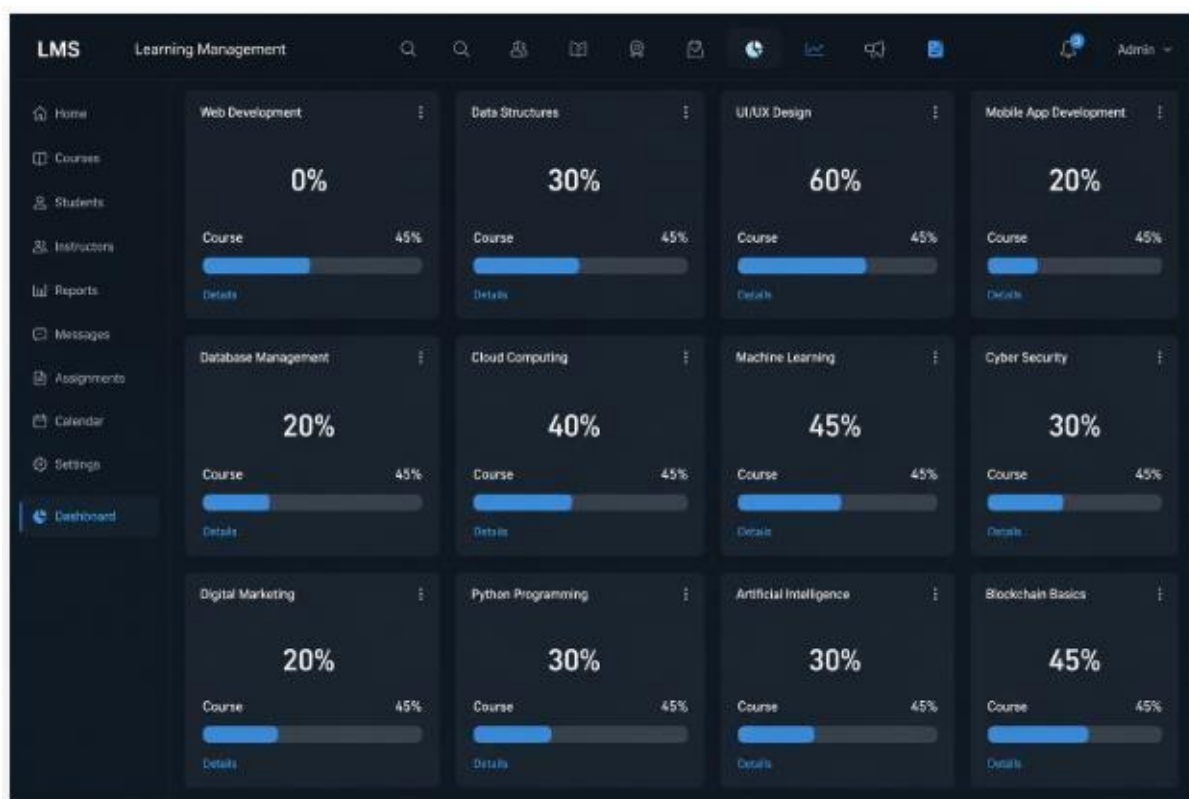


Fig 5.1: Learning Progress Tracker

5.5 Learning Analytics Dashboard

The Learning Analytics Dashboard was studied to understand how learner activity data can be organized into useful metrics. Analytics features help administrator's measure learner engagement, course participation, and completion status. I reviewed dashboard requirements and helped organize information into meaningful sections. The goal was to make learning data easier to interpret.

Working on analytics features showed me that data presentation is as important as data collection. If too much information is shown without structure, users may not understand what action to take. Good analytics dashboards should highlight useful trends, summaries, and indicators. This task improved my understanding of how software can support decision-making.



Fig 5.2: Skill Score Tracker

5.6 Course Recommendation System

The Course Recommendation System introduced me to personalization concepts in learning platforms. The main idea was to suggest relevant courses to learners based on their completed learning path or activity.

Before implementation, I studied possible recommendation criteria and discussed different scenarios with mentors. The objective was to help learners discover useful content without searching manually. I worked on mapping learner activity with available learning content and displaying recommendations on the dashboard. The recommendation logic was kept understandable and practical for the existing system. Different learner profiles were considered to check whether suggestions remained meaningful. This work taught me that recommendation systems should balance simplicity, relevance, and user trust.

5.7 Learning Assistant Chatbot

The final major feature area was the Learning Assistant Chatbot. The purpose of the chatbot was to provide quick support to learners by answering common questions related to courses, progress, deadlines, recommendations, and LMS navigation. I first identified common queries that learners might ask and then planned conversation flows for them.

The chatbot work helped me understand conversational interface design. A chatbot response should be short, clear, and useful. It should guide the learner toward the correct action or section of the LMS. I also studied how chatbot support can reduce the effort required to find information within a platform. This feature showed how user support can be built directly into a software product.

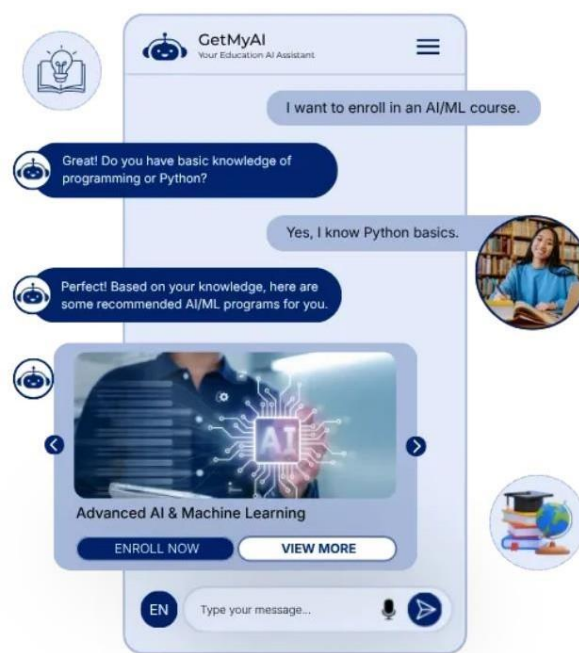


Fig 5.3: Learning Assistance Chatbot

Chapter 6 Results and Discussion

6.1 Summary of Work Completed

The internship resulted in a broad understanding of LMS development and selected practical contributions across frontend and backend areas. The initial result was a properly configured development environment. This allowed local execution and exploration of the project. The next result was a clear understanding of the codebase, module structure, and learner workflows, supported by personal notes and diagrams.

Frontend results included improvements in responsiveness, dashboard organization, form layout review, and visual consistency. Backend learning results included understanding user management, role-based access, course enrollment, progress tracking, analytics preparation, recommendation logic, and chatbot support. The work also resulted in better documentation of the workflows studied during the training period.

6.2 Discussion of Frontend Results

The frontend tasks improved my ability to think from the user perspective. While reviewing dashboard and course management screens, I understood that a good interface should make important actions clear. Responsive testing also showed that a page must be checked under different conditions instead of assuming that it will work everywhere. These lessons are useful for any web application development project.



Fig 6.1: Frontend Contributions

The main value of the frontend work was improvement in usability and consistency. Even when a change was small, it contributed to a better experience for learners or administrators. The work also helped me practice CSS and component-level thinking in a real project, where changes must match existing design patterns.

6.3 Discussion of Backend Results

Backend work helped me understand the logic behind LMS operations. User management introduced the importance of validation and roles. Enrollment showed how different database entities are connected. Progress tracking demonstrated how user activity becomes visible as dashboard information. Analytics showed how raw data can be converted into useful summaries.

The recommendation and chatbot features were especially interesting because they focused on improving learner engagement. Recommendations help users find relevant courses, while chatbot support helps them get quick answers. These features made me understand that modern learning platforms are not only storage systems for content; they also guide and support learners during their journey.

Backend Area	Concept Learned	Practical Importance
User management	Validation, roles, access control	Protects user data and controls permissions
Enrollment	Learner-course relationship	Maintains correct course assignment
Progress tracking	Activity records and completion calculation	Supports learner dashboard and reports
Analytics	Metric preparation and data organization	Helps administrators monitor learning
Recommendation	Rule-based personalization	Improves content discovery
Chatbot	Query handling and response planning	Provides quick learner support

Table 6.1 Backend and database-related learning outcomes

6.4 Challenges Faced

One challenge was understanding a large existing codebase. Unlike small academic projects, an industry project contains many files, shared functions, and dependencies between modules. It took time to understand where a feature was implemented and how data moved through the system. Knowledge transfer sessions and documentation helped me overcome this challenge. Another challenge was making changes without disturbing existing behavior. In a shared project, even a small modification should be checked carefully. I learned to review similar code, test changes locally, and ask for feedback when needed. Responsive design was also challenging because the same screen had to remain usable across multiple device sizes.

6.5 Learning Outcomes

The training improved both my technical and professional skills. Technically, I gained experience with PHP, Git, XAMPP, frontend responsiveness, module-based development, database relationships, analytics concepts, recommendation logic, and chatbot planning. Professionally, I learned the importance of communication, documentation, patience while reading code, and accepting feedback.

The internship also helped me understand the current industry relevance of LMS platforms. Digital learning products are important for organizations because they support continuous training and skill development. By working on such a platform, I could see how software engineering contributes to education and workplace learning.

Chapter 7 Conclusion and Future Scope

7.1 Conclusion

The summer training at Stratbeans Consulting was a practical learning experience. It allowed me to understand how a real learning management system is developed and maintained. The internship strengthened my understanding of web development and software engineering. I learned that a product is not only a collection of code files. It is a complete system created to solve user problems. Every module, whether dashboard, enrollment, progress tracking, recommendation, or chatbot, has a purpose in improving the learning experience. The training also helped me understand the importance of clear communication, documentation, testing, and feedback.

Overall, the internship gave me exposure to industry practices and helped me connect academic learning with practical work. It improved my technical skills in frontend and backend development and also helped me develop professional habits required for future projects.

7.2 Contributions and Utility

My contributions during the internship included studying and documenting LMS workflows, improving selected frontend screens, understanding backend modules, supporting analytics and recommendation-related planning, and preparing chatbot conversation scenarios. The utility of this work lies in improving usability, strengthening documentation, and supporting learner engagement features.

The work also had personal utility because it gave me practical confidence. I became more comfortable with Git workflows, PHP-based development, responsive design, and reading existing code. These skills will help me in future academic projects, internships, and professional work.

7.3 Future Scope

The chatbot can also be expanded to handle more queries and connect with live course data. Future work may include multilingual support, voice-based assistance, and improved integration with course deadlines and notifications. From a development point of view, further improvements can also be made in automated testing, documentation, and performance optimization.

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